Fluid Mechanics

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Types of Fluid Flow

Laminar flow

- Steady flow
- Each particle of the fluid follows a smooth path.
- The paths of the different particles never cross each other.
- Every given fluid particle arriving at a given point has the same velocity.

Turbulent flow

- An irregular flow characterized by small whirlpool-like regions.
- Turbulent flow occurs when the particles go above some critical speed.

Viscosity

- Characterizes the degree of internal friction in the fluid.
- This internal friction, or *viscous force*, is associated with the resistance that two adjacent layers of fluid have to moving relative to each other.
- It causes part of the kinetic energy of a fluid to be converted to internal energy.

Ideal Fluid Flow

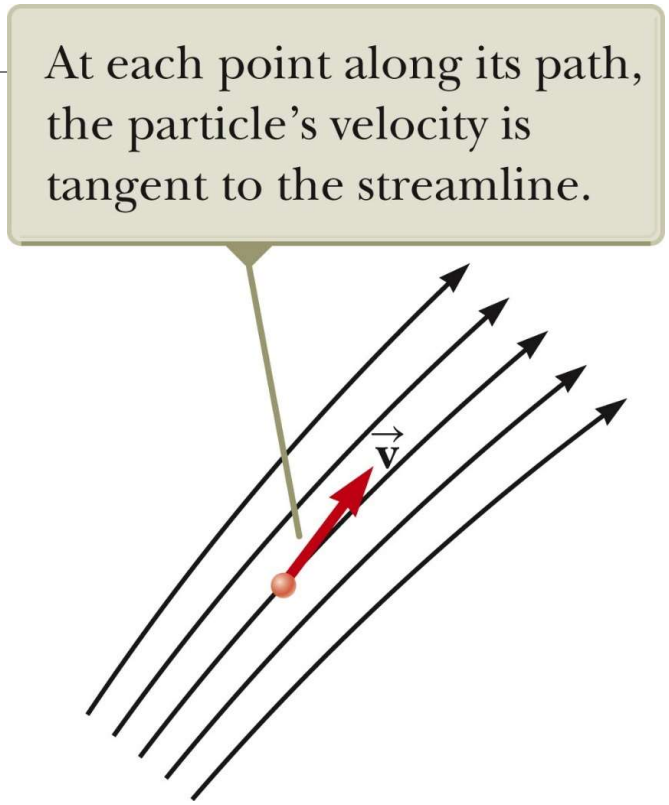
There are four simplifying assumptions made to the complex flow of fluids to make the analysis easier.

- ***The fluid is non-viscous*** – internal friction is neglected
  An object moving through the fluid experiences no viscous forces.
- ***The flow is steady*** – all particles passing through a point have the same velocity.
- ***The fluid is incompressible*** – the density of the incompressible fluid remains constant.
- ***The flow is irrotational*** – the fluid has no angular momentum about any point.

Streamlines

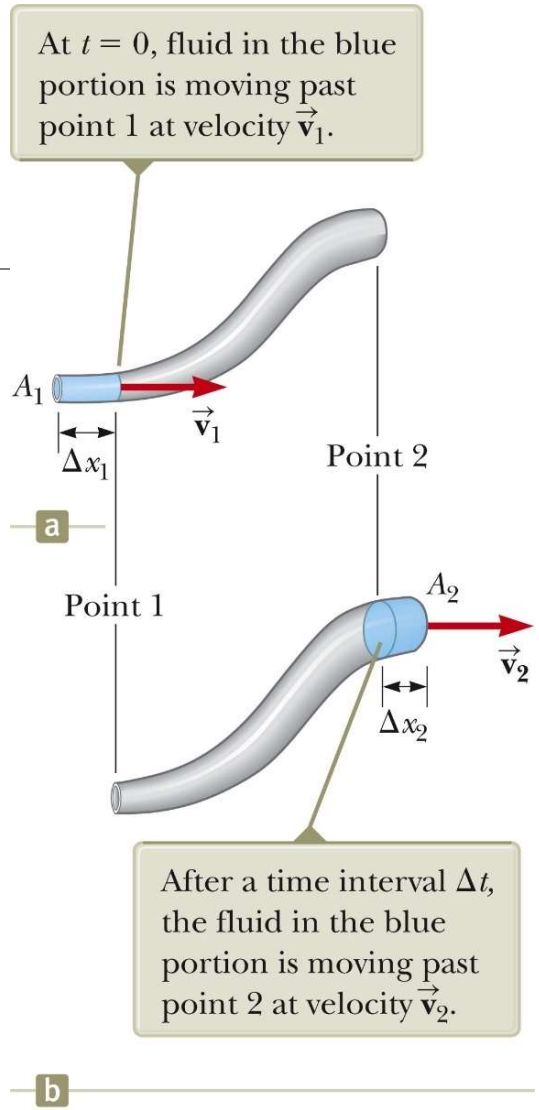
- The path the particle takes in steady flow is a **streamline.**
- The velocity of the particle is tangent to the streamline.
- A set of streamlines is called a *tube of flow.*
- Fluid particles cannot flow into or out of the sides of this tube .

  Otherwise the steamlines would cross each other.



Equation of Continuity

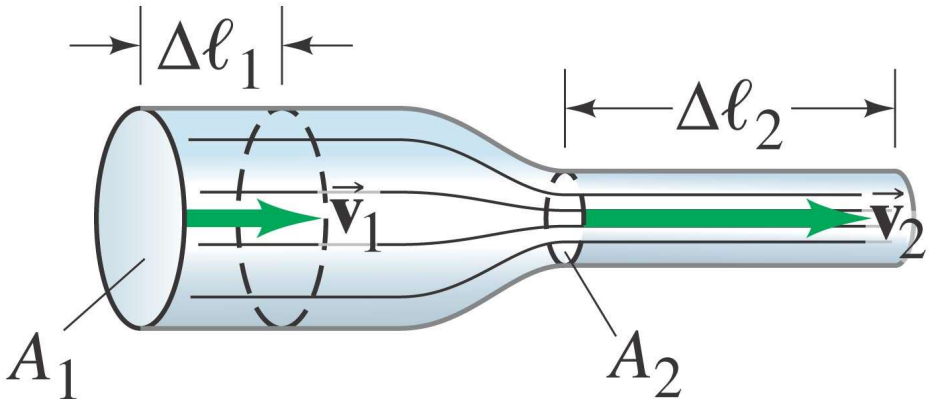
- Consider a fluid moving through a pipe of non-uniform size (diameter).
- Consider the small blue-colored portion of the fluid.
- At t = 0, the blue portion is flowing through a cross section of area A_1 at speed v_1.
- At the end of Δt , the blue portion is flowing through a cross section of area A_2 at speed v_2.
- The mass that crosses A_1 in some time interval is the same as the mass that crosses A_2 in that same time interval.



Equation of Continuity, cont

$$m_1 = m_2$$

$$\rho A_1 v_1 \Delta t = \rho A_2 v_2 \Delta t$$



- The fluid is incompressible,

so ρ is a constant, and flow is constant.

$$A_1 v_1 = A_2 v_2 = constant$$

- This is called the **equation of continuity for fluids.**

Equation of Continuity, Implications

- The speed is high where the tube is constricted (small *A*).
- The speed is low where the tube is wide (large *A*).
- The product, *Av*, is called the *volume flux* or the *flow rate*.
- Av = constant is equivalent to saying the volume that enters one end of the tube in a given time interval equals the volume leaving the other end in the same time. If no leaks are present



George Semple

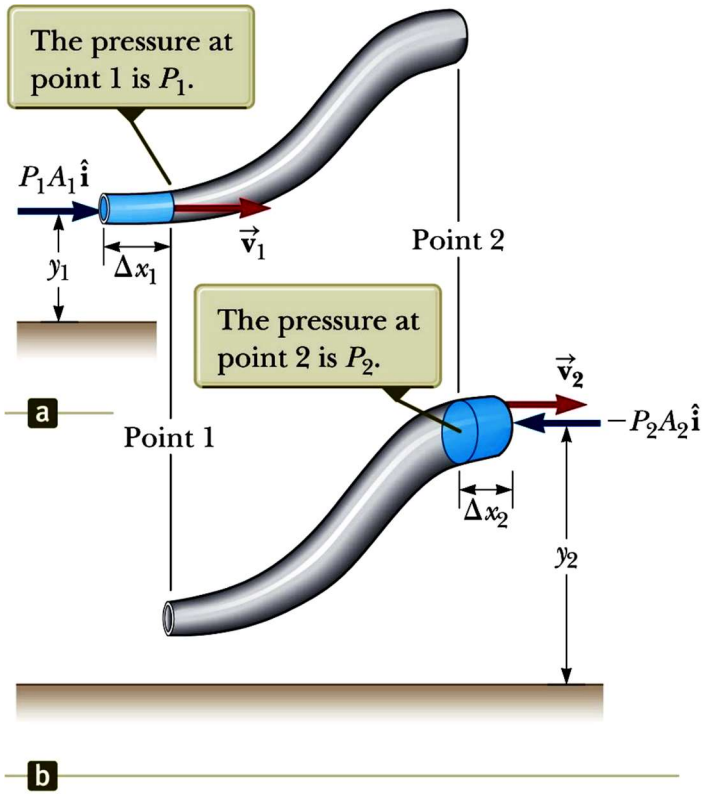
Bernoulli's Equation

- As a fluid moves through a region where its speed and/or elevation above the Earth's surface changes, the pressure in the fluid varies with these changes.

- The relationship between fluid speed, pressure and elevation was first derived by Daniel Bernoulli.

Bernoulli's Equation

- Consider the two shaded segments.
- The volumes of both segments are equal.
- The net work done on the segment is $W = (P_1 - P_2)\, V$.
- Part of the work goes into changing the kinetic energy and some to changing the gravitational potential energy.
- The work is negative because the force on the segment of fluid is to the left and the displacement of the point of application of the force is to the right.

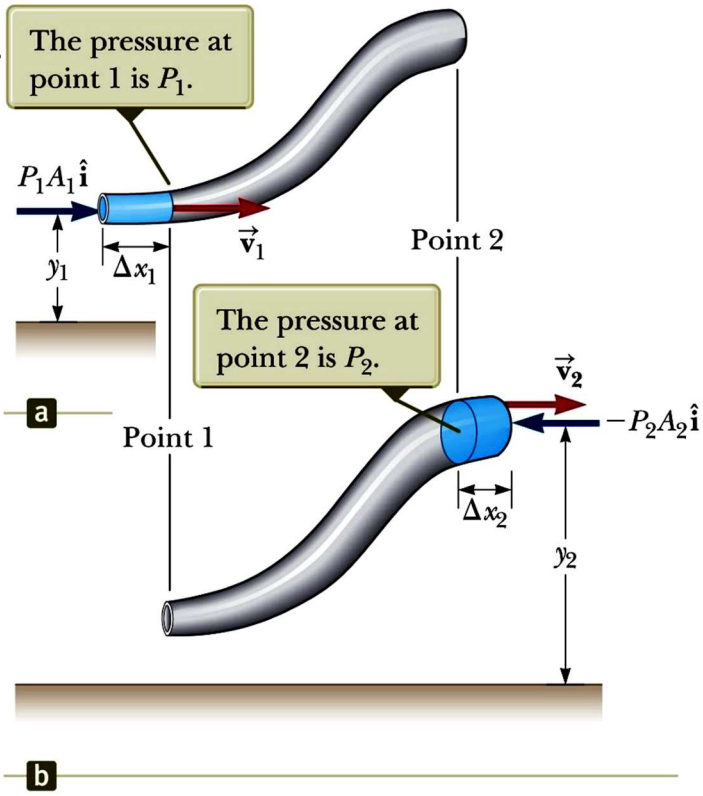


Bernoulli's Equation

Part of the work goes into changing in kinetic energy of the segment of fluid:

$$\Delta K.E. = \tfrac{1}{2}\, mv_2^2 - \tfrac{1}{2}\, mv_1^2$$

- There is no change in the kinetic energy of the gray portion since we are assuming streamline flow.
- The masses are the same since the volumes are the same.



Bernoulli's Equation

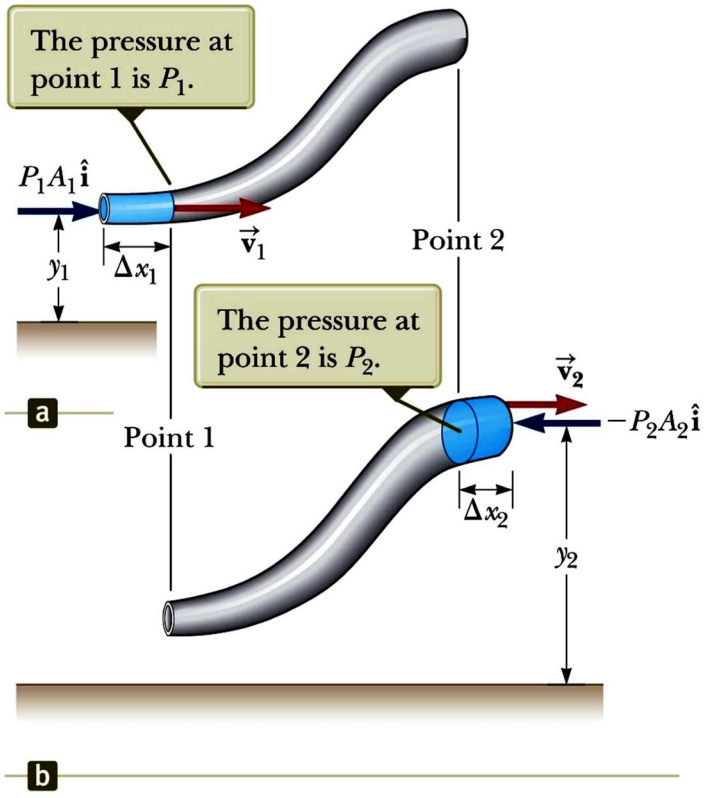
- The change in gravitational potential energy:

$$\Delta U = mgy_2 - mgy_1$$

- The work also equals the change in energy.

Combining:

$$(P_1 - P_2)V = \tfrac{1}{2} mv_2^2 - \tfrac{1}{2} mv_1^2 + mgy_2 - mgy_1$$



Bernoulli's Equation

- Rearranging and expressing in terms of density:

$$P_1 + \tfrac{1}{2}\rho v_1^2 + \rho g y_1 = P_2 + \tfrac{1}{2}\rho v_2^2 + \rho g y_2$$

- This is Bernoulli's Equation as applied to an ideal fluid and is often expressed as

$$P + \tfrac{1}{2}\rho v^2 + \rho g y = constant$$

- When the fluid is at rest, this becomes $P_1 - P_2 = \rho g h$ which is consistent with the pressure variation with depth we found earlier .
- The general behavior of pressure with speed is true even for gases.
  As the speed increases, the pressure decreases.

Applications of Bernoulli's Principle: Torricelli

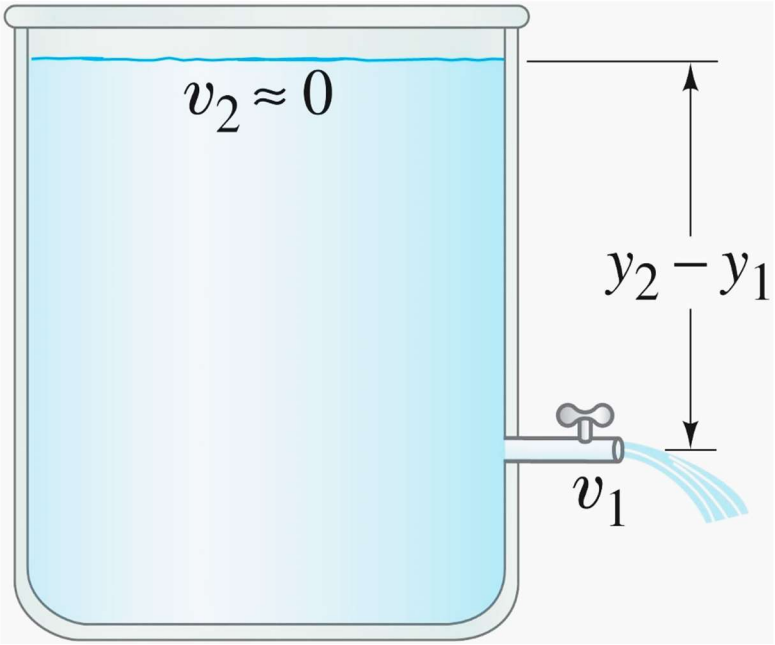
Using Bernoulli's principle, we find that the speed of fluid coming from a spigot on an open tank is:

$$\frac{1}{2}\rho v_1^2 + \rho g y_1 = \rho g y_2$$

or

$$v_1 = \sqrt{2g(y_2 - y_1)}.$$

This is called Torricelli's theorem



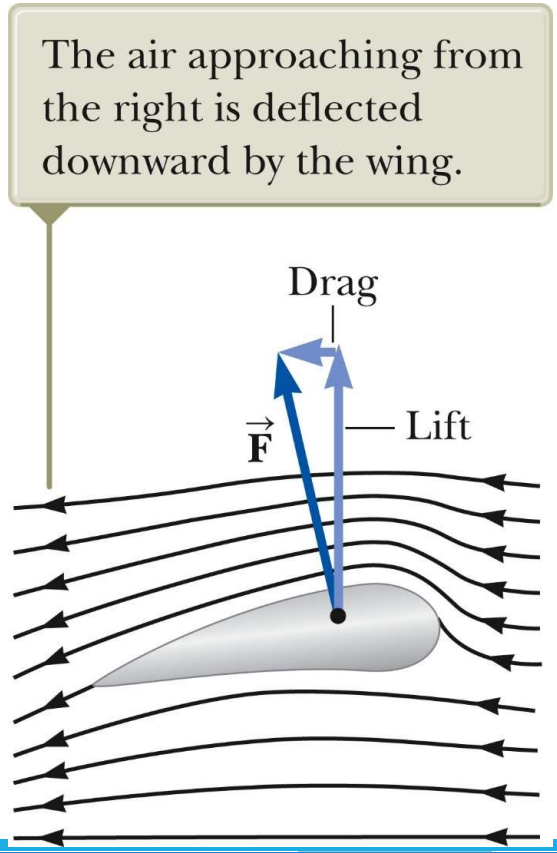
Applications of Bernoulli's Principle: Airplane Wing

Streamline flow around a moving airplane wing.

Lift is the upward force on the wing from the air.

Drag is the resistance.

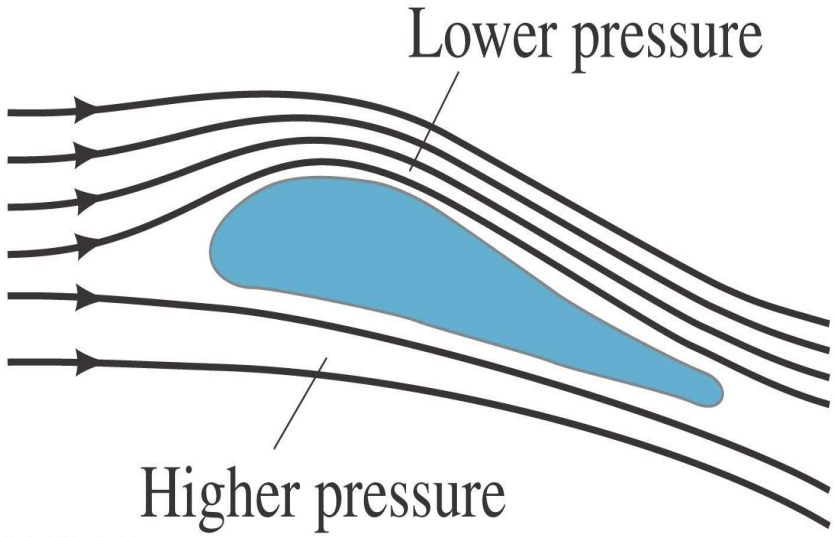
The curvature of the wing surfaces causes the pressure above the wing to be lower than that below the wing due to the Bernoulli effect.



Applications of Bernoulli's Principle: Airplane Wing

- Lift on an airplane wing is due to the different air speeds and pressures on the two surfaces of the wing.

- The lift depends on the speed of the airplane, the area of the wing, its curvature, and the angle between the wing and the horizontal.



Lift – General

In general, an object moving through a fluid experiences lift as a result of any effect that causes the fluid to change its direction as it flows past the object.

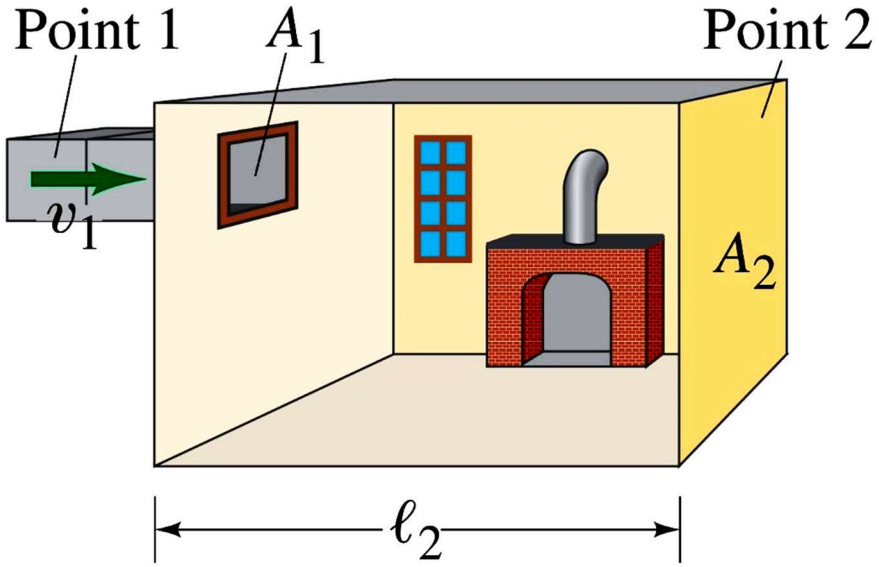
Some factors that influence lift are:

- The shape of the object
- The object's orientation with respect to the fluid flow
- Any spinning of the object
- The texture of the object's surface

Example 1 :

Heating duct to a room.

What area must a heating duct have if air moving 3.0 m/s along it can replenish the air every 15 minutes in a room of volume 300 m^3? Assume the air's density remains constant.



an area of 0.11 m^2.

Example 2 :

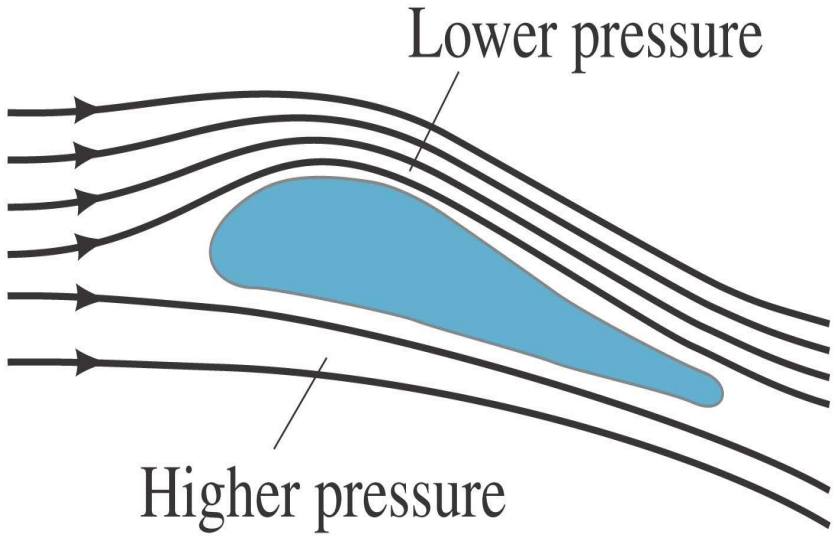
Flow and pressure in a hot-water heating system.

Water circulates throughout a house in a hot-water heating system. If the water is pumped at a speed of 0.5 m/s through a 4.0 cm-diameter pipe in the basement under a pressure of 3.0 atm, what will be the flow speed and pressure in a 2.6-cm-diameter pipe on the second floor 5.0 m above? Assume the pipes do not divide into branches.

The flow speed as 1.2 m/s and the pressure as 2.5 atm.

Example 3 :

An airplane has a mass of 1.60×10^4 kg, and each wing has an area of 40.0 m^2. During level flight, the pressure on the lower wing surface is 7.00×10^4 Pa. Determine the pressure on the upper wing surface.

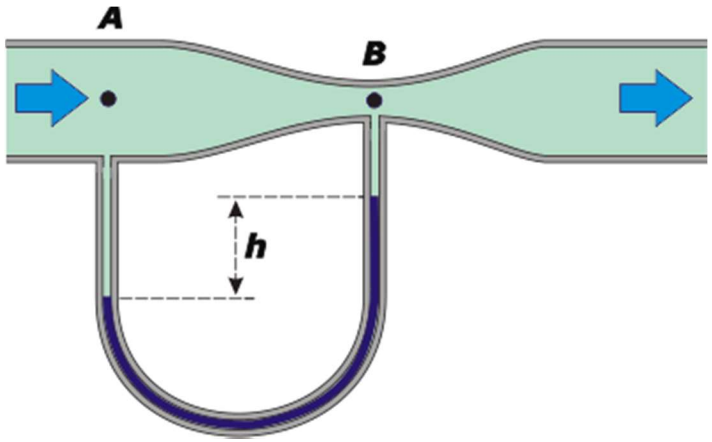


$$\therefore P_2 = 7.0 \times 10^4 - 0.196 \times 10^4 = 6.80 \times 10^4\ Pa$$

Example 4 :

Venturi Tube

For the shown Venturi Tube, determine the velocity of the fluid at the small and large sections of the tube .



Venturi Tube

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g y_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g y_2$$

Since the tube is horizontal

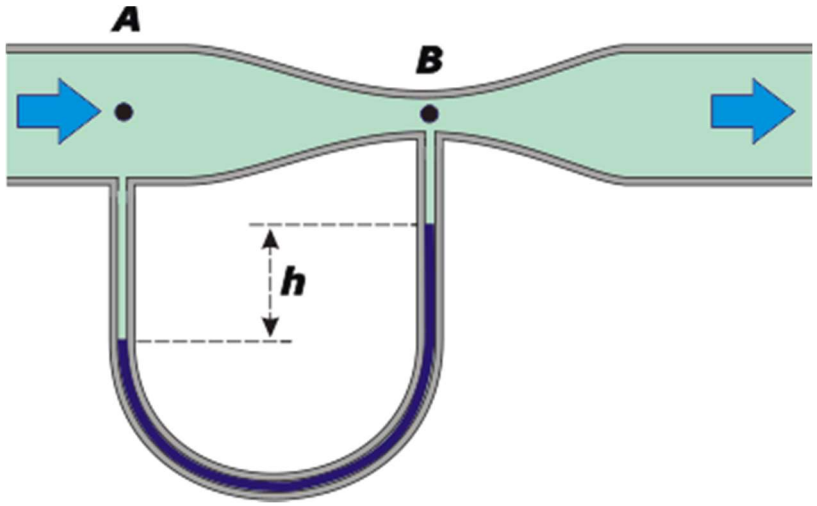
$$\therefore y_2 = y_1$$

$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$$

From the continuity equation:

$$v_1 A_1 = v_2 A_2$$

$$\therefore v_2 = \frac{v_1 A_1}{A_2}$$



Venturi Tube

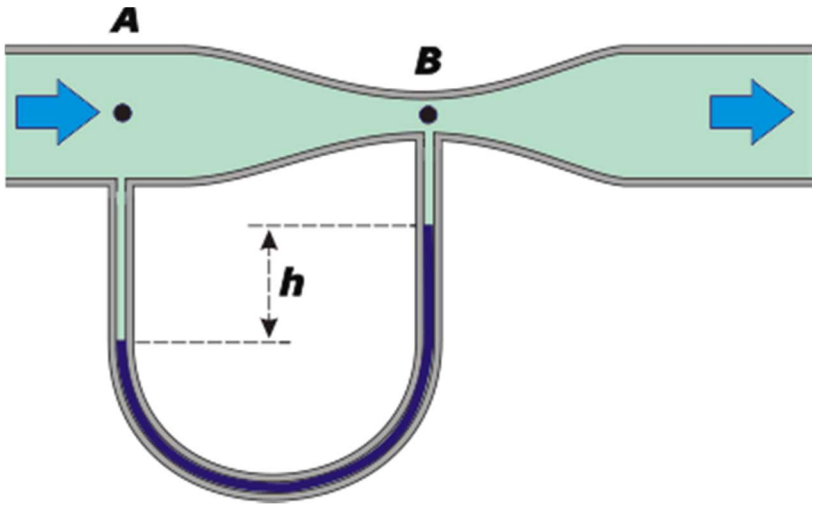
Substituting in Bernoulli's equation

$$\therefore \frac{1}{2}\rho v_1^2\left(\left(\frac{A_1}{A_2}\right)^2-1\right) = P_1 - P_2$$

$$\therefore v_1 = \sqrt{\frac{2(P_1-P_2)}{\rho\left(\left(\frac{A_1}{A_2}\right)^2-1\right)}} = A_2\sqrt{\frac{2(P_1-P_2)}{\rho(A_1^2-A_2^2)}}$$

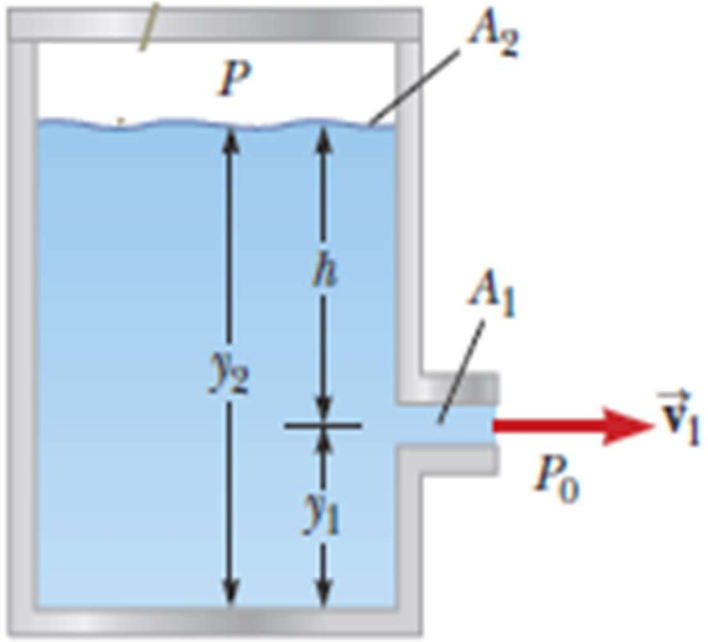
From the continuity equation

$$v_2 = A_1\sqrt{\frac{2(P_1-P_2)}{\rho(A_1^2-A_2^2)}}$$



Example 5 :

An enclosed tank containing a liquid of density ρ has a hole in its side at a distance y_1 from the tank's bottom. The hole is open to the atmosphere, and its diameter is much smaller than the diameter of the tank. The air above the liquid is maintained at a pressure P. Determine the speed of the liquid as it leaves the hole when the liquid's level is a distance h above the hole.



$$v_1 = \sqrt{\frac{2(P - P_0)}{\rho} + 2gh}$$